Yang Yang (3rd Year PhD Candidate)

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Anticipated Graduation Date: 21 May 2028 **Citizenship**: China **Birth**: Sept 2001 **Version**: June 11, 2026
Keywords: Computer Architecture/GPU/Confidential Computing/Cryptography/Performance/Memory Technology

Statement

I am interested in **performance** issues in **GPU** × **Trusted Computing** × **Cryptography** × **Memory**

Recently, I have focused on performance issues in both conventional GPU environments (e.g., multi-GPU system **[ISCA'25]**) and secure GPU environments such as confidential computing (e.g., TDX-based H100 system **[ISPASS'25, ISCA'26]**).

For the next step, I will work on: ***(i) scaling GPU-based CC; (ii) investigating the memory wall problem in GPU-based CC; (iii) making cryptographic protocols GPU-friendly;***

Education

University of Virginia, Charlottesville, USA — **Aug. 2023 – Now**
Ph.D. in Computer Science
GPA: 3.91/4.0
Topics: GPU × {Trusted-Computing, Cryptography, Memory}
Advisor: Dr. Adwait Jog

Jilin University, Changchun, China — **Sept. 2019 – Jul. 2023**
B.S. in Computer Science
GPA: 3.69/4.0
Thesis: The Design and Implementation of Binary Code Analysis Framework for NVIDIA GPU.
Advisor: Dr. Jingweijia Tan

Publications

[1] **(ISCA'26)**
LÆGIS: Pinpointing and Addressing Performance Overheads of GPU-based Confidential Computing
Yang Yang, Adwait Jog
(To Appear) In the Proceedings of IEEE/ACM International Symposium on Computer Architecture, Raleigh, NC, June 2026

[2] **(ISCA'25)**
NetCrafter: Tailoring Network Traffic for Non-Uniform Bandwidth Multi-GPU Systems
Amel Fatima, **Yang Yang**, Yifan Sun, Rachata Ausavarungnirun, Adwait Jog
In the Proceedings of ACM International Symposium on Computer Architecture, Tokyo, Japen, June 2025

[3] **(ISPASS'25)**
Dissecting Performance Overheads of Confidential Computing on GPU-based Systems
Yang Yang, Mohammad Sonji, Adwait Jog
In the Proceedings of IEEE International Symposium on Performance Analysis of Systems and Software, Ghent, Belgium, May 2025

[4] **(SRC@CGO'23)** [poster, before-PhD]
Facilitating Profile Guided Compiler Optimization with Machine Learning
Yang Yang, Xueying Wang, Guangli Li
SRC of IEEE/ACM International Symposium on Code Generation and Optimization, Montreal, Canada, February 2023

Experience

Hardware Security Intern — **May. 2026 – Now**
NVIDIA Corporation, Santa Clara, California, USA
Manager: Vidhya Krishnan, Ambarish Vyas **Mentor**: Santosh Ghosh, Moein Ghaniyoun
Focus: GPU Security, Confidential Computing, Cryptography (Homomorphic Encryption) and LLM Inference

Insight Computer Architecture Lab — **Aug. 2023 – Now**
University of Virginia, Charlottesville, Virginia, USA
Advisor: Dr. Adwait Jog
Focus: Architectural Support for Efficient and Secure GPU Computing.

- GPU-centric memory and storage system (DDR/HBM/SSD with UVM via PCIe/CXL/RDMA). **[ISCA'25, ISCA'26]**
- TDX-based confidential computing on GPUs. **[ISPASS'25, ISCA'26]**
- Cryptography (FHE/PIR) with GPU acceleration (work closely with **Prof. Wei-Kai Lin**).
- Efficient encryption for GPU-based heterogeneous memory system. **[ISCA'26]**

State Key Laboratory of Processor — **Jul. 2022 – Sept. 2023**
Institute of Computing Technology, Chinese Academy of Science, Beijing, China
Advisor: Dr. Guangli Li
Topics: Compiler & Programming Systems
Focus: Facilitating Profile-Guided Compiler Optimization with Graph Learning

- Proposed a branch predictor using XGBoost based on compile time static analysis. **[CGO-SRC'23]**
- Utilize graph representations to build predictive profile-guided optimization framework and integrate it into LLVM.

Emerging Technology Enabled Computer Architecture Lab — **Feb. 2022 – Jul. 2023**
Jilin University, Changchun, Jilin, China
Advisor: Dr. Jingweijia Tan
Topics: GPU × {PTX/SASS, Reliability, Energy Efficiency}
Focus: Reliability and Energy-Efficiency of GPUs

- Soft error and hard error (e.g., process variation, NBTI) on FinFET based chiplet GPUs.
- SASS level analysing and programming framework for NVIDIA Ampere GPUs. [**Thesis**]
- Deep learning techniques for GPU power modeling.
- Explored how to reverse-engineer SASS and program in SASS.
- Instruction level under-voltage reliability of GPUs

Talks

LÆGIS: Pinpointing and Addressing Performance Overheads of GPU-based Confidential Computing

- UVA CS SIG. May 2026

NetCrafter: Tailoring Network Traffic for Non-Uniform Bandwidth Multi-GPU Systems

- ACM ISCA. June 2025
- UVA CS SIG. September 2025

Dissecting Performance Overheads of Confidential Computing on GPU-based Systems

- IEEE ISPASS. May 2025
- UVA CS SIG. May 2025
- IBM Research (Virtual). May 2025
- Confidential Computing Summit. June 2025

Teaching

25 Fall @ UVA, TA for CS 4444: Introduction to Parallel Computing (undergraduate)

- Co-Designed CUDA programming assignments.
- Delivered a lecture on GPU-based confidential computing.

24 Fall @ UVA, TA for CS 6354: Computer Architecture (graduate)

- Designed a pipelined RISC-V simulator in Python.

Award

[**Travel Grant**] ISPASS'25 ISCA'25 ISCA'26

[**Fellowship@UVA**] 2023

[**Scholarship@JLU**] 2019 2020 2021 2022

Service

[**Submission Chair**] IISWC'26

[**GPU-RIG**] Coordinating logistics for GPU Research Interest Group across UVA SEAS in 2026.

[**Artifact Evaluation Committee**] IISWC'25

[**SIG**] Co-organizer of Systems-Interest-Group (SIG) meetings at UVA Department of Computer Science in 2024 and 2025.

Skills

[**Languages**] C/C++, Python, Assembly, Go, Rust

[**Frameworks**] CUDA, Pytorch, XGBoost, LLVM, TDX/SGX, QEMU/KVM, NVIDIA GPU Kernel Module, OpenFHE

[**Software**] Linux, LaTeX, GNU compilers, GPGPU-Sim/Accel-Sim, NVCC/NCU/NSYS/NVBit, DRAMSim, Varius-TC, Z3

[**Cryptography**] AES (GCM, SIV, *etc.*), Merkle Trees, Public-Key (DH, RSA, *etc.*), PIR (LWE-based, FHE-based), ORAM

PhD Committee (Qual Exam)

Chair: Wei-Kai Lin **Members**: Sandhya Dwarkadas Felix Lin Adwait Jog **Announcement:** [Event link]